192425192

G Vishnu Madhav

EXP5:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int gcd(int a, int b) { return b == 0 ? a : gcd(b, a % b); }

int modinv(int a, int m) {
    int x;
    for (x = 1; x < m; x++)
        if ((a * x) % m == 1) return x;
    return -1;
}

int main() {
    char text[200], enc[200], dec[200];
    int a, b, i, ainv;

    printf("Values of a allowed (gcd(a,26)=1): ");
    for (a = 1; a < 26; a++)
        if (gcd(a, 26) == 1) printf("%d ", a);
    printf("\nValues of a NOT allowed: ");
    for (a = 1; a < 26; a++)
        if (gcd(a, 26) != 1) printf("%d ", a);
    printf("\nb can be any value from 0 to 25 (no restriction).\n\n");

    printf("Enter plaintext: ");
    scanf("%s", text);
```

```
    printf("Enter a and b: ");
    scanf("%d %d", &a, &b);

    if (gcd(a, 26) != 1) {
        printf("Invalid key: a=%d is not coprime with 26, cipher is not one-to-one.\n", a);
        return 0;
    }
    ainv = modinv(a, 26);
    for (i = 0; text[i]; i++) {
        int p = toupper(text[i]) - 'A';
        enc[i] = (a * p + b) % 26 + 'A';
    }
    enc[i] = '\0';
    for (i = 0; enc[i]; i++) {
        int c = enc[i] - 'A';
        dec[i] = (ainv * ((c - b + 26) % 26)) % 26 + 'A';
    }
    dec[i] = '\0';
    printf("Ciphertext: %s\n", enc);
    printf("Decrypted : %s\n", dec);
    return 0;
}
```

OUTPUT:

```
Values of a allowed (gcd(a,26)=1): 1 3 5 7 9 11 15 17 19 21 23 25
Values of a NOT allowed: 2 4 6 8 10 12 13 14 16 18 20 22 24
b can be any value from 0 to 25 (no restriction).

Enter plaintext: HELLOWORLD
Enter a and b: 5 8
Ciphertext: RCLLAOAPLX
Decrypted : HELLOWORLD
```